Returns to Education in Canada

An Econometric Analysis of Real Wages, 2018–2025

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KEY FINDINGS

- A Bachelor's degree carries a +26.8% wage premium over 0–8 years of schooling, and an above-bachelor's credential +34.9% — both significant at the 1% level, holding occupation, industry, region, and demographics fixed.
- The gender wage gap is not explained by education, occupation, or industry: women earn 15.7% less than men after controlling for all observed characteristics.
- Industry matters as much as credentials. Finance & Insurance carries a +30% wage premium over the reference sector; Mining/Oil/Gas +50%. Two workers with identical education and experience can earn very differently based on sector alone.
- Regional pay gaps persist after controls: Alberta (+12.4%) and British Columbia (+11.1%) pay a premium; Atlantic provinces carry a 5–7% penalty.

Motivation

Education is one of the most-studied determinants of earnings in labour economics, but the size of the return varies by country, labour market structure, and time period — and Canadian estimates using recent, post-pandemic data are scarce. This paper estimates the wage premium associated with each level of educational attainment in Canada, using eight years of pooled Labour Force Survey microdata and a regression specification that controls for the confounders most likely to bias a naive comparison: occupation, industry, union status, firm size, family structure, province, and year.

Prior work establishes the direction of the effect. Mincer (1974) is the foundational reference but is based on U.S. data. Oreopoulos (2006) exploits Canadian compulsory schooling law as a natural experiment and finds a genuine causal effect, but only for compulsory schooling — it says nothing about the postsecondary margin most relevant to today's labour market. Lemieux (2010) documents that Canadian wage structure has shifted meaningfully by field of study, gender, and geography. This paper updates the estimate with current data and asks not just whether education pays, but how much of the answer depends on where and in what you work.

Data & Methodology

The sample is drawn from Statistics Canada's Labour Force Survey Public Use Microdata File (PUMF), pooled 2018–2025 and restricted to employed individuals aged 20+ with a valid hourly wage. After restrictions, the estimation sample contains 1,759,743 observations (≈558M weighted person-records). Survey weights are applied throughout; standard errors are clustered at the province level (10 clusters).

The dependent variable is the natural log of real hourly wage (September 2025 dollars). The model is a fully saturated Mincer-style specification:

$$ln(w_i) = \beta_1 Education_i + X_i'\gamma + \delta_p + \delta_t + \varepsilon_i$$

Education enters as a seven-category indicator (reference: 0–8 years of schooling). X_i includes gender, age, marital status, family type, occupation (8 major groups), industry (18 sectors), union status, establishment and firm size, and multiple-jobholding status. δ_p and δ_t are province and year fixed effects. The full specification, Stata command, and complete coefficient table for every control variable are in the Appendix.

Before turning to the regression, the unconditional wage distributions are worth inspecting directly. A kernel density plot of log wages for workers with 0–8 years of schooling versus Bachelor's-degree holders shows the university-educated distribution isn't just shifted right — it's also more dispersed, with a materially thicker upper tail. Education raises the average and widens the range of outcomes.

Results

Education Returns

Returns to education are strong and monotonic across every category:

Education Level (vs. 0–8 yrs)	Wage Premium	p-value	Sig.
Some high school	+3.8%	<0.001	***
High school graduate	+6.7%	<0.001	***
Some postsecondary	+10.1%	<0.001	***
Postsecondary diploma	+14.1%	<0.001	***
Bachelor's degree	+26.8%	<0.001	***
Above bachelor's degree	+34.9%	<0.001	***

All estimates are significant at the 1% level, adjusted for province clustering. The premium accelerates sharply above the postsecondary-diploma margin — the jump from diploma to Bachelor's (+14.1% to +26.8%) is nearly twice the jump from high school to diploma (+6.7% to +14.1%).

What Education Doesn't Explain

Three findings condition how much weight the headline number should carry on its own:

- Gender: Women earn 15.7% less than men, controlling for education, occupation, industry, region, and job characteristics — consistent with Statistics Canada's own decomposition work on the unexplained portion of the gap.
- Occupation: Non-management roles carry wage penalties of −20% to −58% relative to management, even within the same education and industry cell. Manufacturing/Utilities and Trades/Transport occupations see the steepest discounts.
- Industry: Wage premia range from +50% (Mining/Oil/Gas) and +30% (Finance & Insurance) down to statistically indistinguishable from zero (Retail, Accommodation/Food). Two otherwise-identical workers can see a 30-40 point difference in real wages based on sector choice alone.

Regional effects are also large and persistent: Alberta (+12.4%) and British Columbia (+11.1%) carry premiums; Atlantic Canada carries a 5–7% penalty — even after conditioning on the industry and occupation mix specific to each region.

Limitations

This is a cross-sectional, not causal, estimate — the education coefficients reflect conditional correlation, not a clean identification strategy (no instrument or natural experiment is used, unlike Oreopoulos 2006). Selection into education is not addressed: individuals with higher unobserved ability or motivation may both pursue more schooling and earn more for reasons unrelated to the credential itself, which would bias the coefficients upward. The LFS PUMF also caps top-coded wage values and geographic detail is limited to CMA/province, which understates within-region heterogeneity. These estimates should be read as the wage gap associated with education level in the current labour market, not the causal return an individual could expect from an additional credential.

Why It Matters

The finding that industry and occupation carry premiums comparable in size to an entire additional credential — and that a well-documented, unexplained gender gap survives every control available in a national dataset — is the more actionable result for anyone using this kind of analysis to inform capital allocation, workforce planning, or compensation benchmarking. Credentials matter, but where and in what role someone works can matter just as much or more, and that's a testable, monitorable fact rather than a slow-moving policy lever.

Appendix

Full regression output (all control variables), Stata command, and figure-generation code are available on request. Summary: $R^2 = 0.484$; n = 1,759,743; robust SEs clustered by province (10 clusters).

Regression command: reg ln_real_wage i.education experience i.GENDER i.marital_status i.province i.cma i.union_member i.occupation_major i.industry i.establishment_size i.school_attendance i.economic_family_type i.age_group5 i.multiple_jobs i.class_of_worker i.age_youngest_child i.firm_size i.year [aw=weight], vce(cluster province)

References

Becker, G. S. (1964). Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education.

Lemieux, T. (2010). What do we really know about changes in wage inequality. NBER Chapters, 17–59.

Mincer, J. A. (1974). Schooling, experience, and earnings.

Oreopoulos, P. (2006). Estimating Average and Local Average Treatment Effects of Education when Compulsory Schooling Laws Really Matter. American Economic Review, 96(1), 152–175.